NEET UG 2027: Master Biology Chapter Notes

Reproduction in Plants & Humans, Reproductive Health & Biotechnology

Prepared, Compiled & Authored by Satya Prakash · Strictly Calibrated to Rationalized NCERT & NMC Scope

1 Sexual Reproduction in Flowering Plants (Angiosperms)

Microsporogenesis & Male Gametophyte (Pollen Grain)

- **Anther Structure:** Typically bilobed, dithecous (two thecae per lobe), and tetrasporangiate (four microsporangia).
- **Wall Layers (from outside to inside):**
  1. **Epidermis:** Protective single outer layer.
  2. **Endothecium:** Layer with fibrous hygroscopic α-cellulose bands; assists in anther dehiscence.
  3. **Middle Layers:** $1-3$ transient ephemeral layers; degenerate to provide nourishment.
  4. **Tapetum:** Innermost nutritive layer; cells possess dense cytoplasm and are generally **multinucleate / polyploid**; secretes **sporopollenin**, **callase**, and pollenkitt.
- **Pollen Grain Wall (Sporoderm):**
  - **Exine:** Hard outer layer composed of **Sporopollenin** (the most resistant organic substance known; withstands high temperatures, strong acids, and alkali; no enzyme that degrades sporopollenin is known). Has apertures called **Germ Pores** where sporopollenin is absent.
  - **Intine:** Thin, continuous inner wall composed of **cellulose and pectin**.
- **Pollen Shedding Stage:** In $> 60\%$ of angiosperms, pollen is shed at the **2-celled stage** (large vegetative cell with irregular nucleus + small spindle-shaped generative cell floating in vegetative cytoplasm). In $< 40\%$, generative cell divides mitotically to form 2 male gametes before shedding (**3-celled stage**).

Megasporogenesis & The Monosporic Polygonum-Type Embryo Sac

- **Ovule (Integumented Megasporangium):** Attached to placenta by **funicle**; point of attachment to ovule body is **hilum**. Massive central tissue is **nucellus**.
- **Megasporogenesis:** Megaspore Mother Cell (MMC) undergoes meiosis to produce a linear tetrad of 4 haploid megaspores. 3 micropylar megaspores degenerate; only **1 chalazal megaspore remains functional** (**Monosporic development**).
- **Embryo Sac Organization (7-celled, 8-nucleate at maturity):**
  - **Micropylar End (Egg Apparatus - 3 cells):** 1 central **Egg cell** + 2 flanking **Synergids** possessing **Filiform apparatus** (finger-like wall projections that guide pollen tube entry).
  - **Chalazal End (Antipodals - 3 cells):** Vegetative nutritive cells; degenerate after fertilization.
  - **Central Cell (1 large cell):** Contains **2 Polar Nuclei** that fuse before fertilization to form a diploid **Secondary Nucleus** ($2n$).

Double Fertilization, Endosperm & Apomixis

1. Double Fertilization (Discovered by S.G. Nawaschin, 1898): Unique to angiosperms; involves two distinct nuclear fusion events:

$$\begin{aligned} \textbf{Syngamy:}\ & \text{Male Gamete } (n) + \text{Egg } (n) \longrightarrow \text{Zygote } (\mathbf{2n}) \longrightarrow \text{Embryo} \\ \textbf{Triple Fusion:}\ & \text{Male Gamete } (n) + \text{Secondary Nucleus } (2n) \longrightarrow \text{PEN } (\mathbf{3n}) \longrightarrow \text{Endosperm} \\ \textbf{Total Nuclei:}\ & 1\ (\text{sperm}) + 1\ (\text{egg}) + 1\ (\text{sperm}) + 2\ (\text{polar nuclei}) = \mathbf{5}\ \text{Nuclei} \end{aligned}$$

2. Endosperm Development: Triploid Primary Endosperm Nucleus (PEN) divides to form endosperm before the zygote divides (ensures nutrition for embryo).

- **Free-Nuclear Endosperm:** PEN undergoes repeated free nuclear divisions without cell wall formation

(e.g., coconut water is free-nuclear endosperm consisting of thousands of free nuclei; surrounding white kernel is cellular endosperm).

3. **Special Reproductive Modes:**

- **Apomixis:** Form of asexual reproduction that **mimics sexual reproduction**, producing seeds without fertilization (e.g., *Asteraceae* and grasses). Nucellar cells divide and protrude into embryo sac to form diploid embryos.
- **Polyembryony:** Occurrence of more than one embryo in a seed (common in *Citrus* and mango).

2 Human Reproduction & Reproductive Health

Male & Female Reproductive Architecture

1. **Male System:**

- **Scrotum:** Keeps testes at $2-2.5^{\circ}$C lower than internal core body temperature, necessary for spermatogenesis.
- **Seminiferous Tubules:** Lined by (i) **Spermatogonia** (male germ cells) and (ii) **Sertoli cells** (nurse cells providing nutrition, secrete Androgen Binding Protein and Inhibin).
- **Interstitial / Leydig Cells:** Located in interstitial spaces outside tubules; synthesize and secrete androgens (**Testosterone**) under LH stimulation.
- **Accessory Glands:** Paired Seminal vesicles (contribute $60-70\%$ of semen, rich in **fructose, calcium**, and prostaglandins), single Prostate gland (milky alkaline fluid), paired Bulbourethral / Cowper's glands (secrete alkaline mucus for lubricating penis).

2. **Female System:**

- **Ovaries:** Primary female sex organs producing ovum and steroid hormones (Estrogen, Progesterone).
- **Fallopian Tube (Oviduct, 10–12 cm):** Consists of (i) **Infundibulum** with finger-like **fimbriae** to collect ovum after ovulation, (ii) **Ampulla** (the wider part; **site of fertilization**), (iii) **Isthmus** (narrow lumen joining uterus).
- **Uterine Wall:** Outer thin **perimetrium**, middle thick smooth-muscle **myometrium** (exhibits strong contractions during parturition), and inner glandular **endometrium** (undergoes cyclic changes during menstrual cycle).

Gametogenesis: Spermatogenesis vs Oogenesis Timing

1. **Spermatogenesis (Continuous from Puberty through Old Age):**

$$\text{Spermatogonia } (2n) \xrightarrow{\text{mitosis}} \text{Primary Spermatocyte } (2n) \xrightarrow{\text{Meiosis I}} 2 \times \text{Secondary Spermatocytes } (n)$$
$$\xrightarrow{\text{Meiosis II}} 4 \times \text{Spermatids } (n) \xrightarrow{\textbf{Spermiogenesis}} 4 \times \text{Spermatozoa } (\mathbf{n})$$

2. **Oogenesis (Initiated during Fetal Life, Suspended until Puberty):**

$$\text{Oogonia } (2n) \xrightarrow{\text{mitosis}} \text{Primary Oocyte } (2n) \xrightarrow{\text{Arrested at Prophase I}} \text{At Birth}$$
$$\text{At Puberty: Completes Meiosis I} \longrightarrow \text{Secondary Oocyte } (\mathbf{n}) + \mathbf{1}^{\text{st}} \text{ Polar Body } (\mathbf{n})$$
$$\text{Fertilization: Completes Meiosis II} \longrightarrow \text{Ovum } (\mathbf{n}) + \mathbf{2}^{\text{nd}} \text{ Polar Body } (\mathbf{n})$$

Menstrual Cycle Hormonal Regulation & Implantation

Standard 28-day cycle divided into four phases:

1. **Menstrual Phase (Days 1–5):** Breakdown of endometrial lining and blood vessels due to sudden **withdrawal of progesterone**; average blood loss $= 50-100$ mL.
2. **Follicular / Proliferative Phase (Days 6–13):** Pituitary **FSH** stimulates maturation of primary follicle into mature **Graafian follicle**; growing follicle secretes **Estrogen**, which regenerates endometrium.
3. **Ovulatory Phase (Day 14):** Peak levels of both LH and FSH; rapid surge of LH (**LH Surge**) induces rupture of Graafian follicle and release of secondary oocyte (**Ovulation**).

4. **Luteal / Secretory Phase (Days 15–28):** Ruptured follicle transforms into endocrine **Corpus Luteum**, which secretes massive amounts of **Progesterone** (essential for maintenance of endometrium). If fertilization fails, corpus luteum degenerates into **Corpus Albicans**, causing progesterone drop and menses.

Fertilization & Pregnancy Endocrinology:

- **Site of Fertilization:** Ampullary-isthmic junction / **Ampulla** of Fallopian tube.
- **Cortical Reaction:** Sperm induces depolarization and hardening of **zona pellucida**, blocking polyspermy.
- **Implantation:** Trophoblast layer of **Blastocyst** attaches to uterine endometrium ≈ 7 days after fertilization.
- **Placental Hormones exclusively in pregnancy:** Human Chorionic Gonadotropin (**hCG**), Human Placental Lactogen (**hPL**), and **Relaxin** (secreted by ovary in late pregnancy).

Reproductive Health: Contraception, MTP & ART Protocols

1. Contraceptive Methodologies:

- **Natural / Traditional:** Periodic abstinence (avoid coitus days $10 - 17$ of cycle), coitus interruptus, **lactational amenorrhea** (intense lactation prevents ovulation for max 6 months postpartum).
- **Barrier Methods:** Condoms (protect against STIs/AIDS), Diaphragms, cervical caps, vaults.
- **Intrauterine Devices (IUDs):**
  - **Non-medicated IUDs:** *Lippes loop* (increases phagocytosis of sperms).
  - **Copper-releasing IUDs: CuT, Cu7, Multiload 375** (Cu^{2+} ions suppress sperm motility and fertilizing capacity).
  - **Hormone-releasing IUDs: Progestasert, LNG-20** (make uterus unsuitable for implantation and cervix hostile to sperms).
- **Oral Contraceptive Pills:** Daily combined progestogen-estrogen pills; or **Saheli** (developed by CDRI Lucknow; non-steroidal, once-a-week pill with high contraceptive value and negligible side effects).
- **Surgical / Permanent Methods: Vasectomy** (cutting/tying vas deferens in males) and **Tubectomy** (fallopian tubes in females); irreversible sterilization.

2. Assisted Reproductive Technologies (ART):

- **IVF-ET (In Vitro Fertilization - Embryo Transfer):** Test-tube baby program.
  - **ZIFT (Zygote Intra-Fallopian Transfer):** Zygote or early embryo **up to 8 blastomeres** transferred into Fallopian tube.
  - **IUT (Intra-Uterine Transfer):** Embryos with **more than 8 blastomeres** transferred directly into Uterus.
- **GIFT (Gamete Intra-Fallopian Transfer):** Transfer of ovum collected from donor into fallopian tube of female who cannot produce one.
- **ICSI (Intra-Cytoplasmic Sperm Injection):** Sperm directly injected into ovum cytoplasm in laboratory.
- **IUI (Intra-Uterine Insemination):** Semen collected from husband/donor introduced artificially into uterus.

3 Biotechnology: Principles, Tools & Modern Applications

Tools of Recombinant DNA Technology

1. Restriction Endonucleases ("Molecular Scissors"): Discovered by Arber, Smith, and Nathans (1970). Cut double-stranded DNA at specific palindromic recognition sequences.

- **First Discovered:** *Hind II* (cuts at specific 6 bp sequence).
- **Nomenclature (*Eco*RI):** *E* for genus *Escherichia*, *co* for species *coli*, R for strain RY13, I for first enzyme isolated.

- **Recognition Sequence of *Eco*RI:**

$$5' - \text{G} \downarrow \text{A A T T C} - 3' \qquad 3' - \text{C T T A A} \uparrow \text{G} - 5' \tag{1}$$

Produces staggered single-stranded overhangs called **sticky ends**.

2. Cloning Vectors (Features of Ideal Vector pBR322):

- **Origin of Replication (*Ori*):** Specific sequence where replication initiates; controls copy number of linked foreign DNA.
- **Selectable Markers:** Identify and select transformants and eliminate non-transformants. Example: genes conferring resistance to antibiotics (amp^R **and** tet^R).
- **Insertional Inactivation:** Ligation of foreign DNA at *BamHI* site of tet^R gene inactivates tetracycline resistance $\implies$ recombinant grows on ampicillin, dies on tetracycline.
- **Blue-White Screening:** Foreign DNA inserted into gene coding for enzyme **β-galactosidase** (α-complementation). Non-recombinants produce blue colonies with X-gal; recombinants produce **white colonies**.

3. Competent Host Transformation Techniques:

- **Chemical Method:** Treating bacterial cells with divalent cation (Ca^{2+}) followed by **Heat Shock** (42°C for 2 minutes and ice plunge).
- **Microinjection:** Recombinant DNA directly injected into nucleus of animal cell.
- **Biolistics / Gene Gun:** Plant cells bombarded with high-velocity microparticles of **gold or tungsten** coated with DNA.
- **Disarmed Pathogen Vectors:** *Agrobacterium tumefaciens* (Ti plasmid for dicots) and retroviruses for animal cells.

Polymerase Chain Reaction (PCR) & Downstream Processing

Invented by Kary Mullis in 1983. In vitro enzymatic amplification of DNA sequence by a billion-fold (2^n copies):

Step 1: Denaturation: $94 - 96°\text{C}$ (Thermal separation of dsDNA strands by breaking H-bonds)
Step 2: Annealing: $50 - 60°\text{C}$ (Hybridization of two oligonucleotide primers to flanking $3'$-ends)
Step 3: Extension: $72°\text{C}$ (Extension by Taq Polymerase from *Thermus aquaticus*)

Bioreactors & Downstream Processing:

- **Bioreactors ($100 - 1000\,\text{L}$):** Provide optimal growth conditions (pH, temperature, oxygen, substrates, salts). **Stirred-tank** provides continuous mixing; **sparged stirred-tank** bubbles sterile air through broth.
- **Downstream Processing:** Separation and purification of biosynthesized product, followed by clinical trials, formulation with preservatives, and stringent quality control.

High-Yield Transgenic Applications & Medical Biotechnology

1. **Bt Cotton & Pest Resistance:** Soil bacterium *Bacillus thuringiensis* produces crystalline proteinaceous protoxins (**Cry proteins**). The inactive protoxin is converted into active toxin in the **alkaline pH of the insect midgut**, binding to epithelial cells, creating pores, causing cell lysis and insect death:

CryIAc and **CryIIAb** control Cotton Bollworms **CryIAb** controls Corn Borer (2)

2. **RNA Interference (RNAi):** Cellular defense mechanism in all eukaryotes. Nematode *Meloidogyne incognita* infects roots of tobacco plants. Using *Agrobacterium* vectors, nematode-specific genes are introduced producing both sense and anti-sense RNA $\implies$ forms **double-stranded RNA (dsRNA)** $\implies$ silences specific mRNA of nematode.
3. **Genetically Engineered Insulin (Humulin):** Human insulin consists of chain A (21 amino acids) and chain B (30 amino acids) linked by **disulphide bonds**. Synthesized as **prohormone containing an extra C-peptide** (33 amino acids), which is removed during maturation. In 1983, American company **Eli Lilly** produced two separate DNA sequences for chain A and chain B, introduced into *E. coli*, and combined them by creating disulphide bonds.
4. **Gene Therapy for ADA Deficiency:** First clinical gene therapy performed in 1990 on a 4-year-old girl

with **Adenosine Deaminase (ADA) deficiency** (causes Severe Combined Immunodeficiency - SCID). Lymphocytes from patient's blood transduced with functional ADA cDNA using a retroviral vector and returned to patient. (Requires periodic infusion unless embryonic stem cells are engineered).

5. **Transgenic Animals & Biopiracy:**
   - **First Transgenic Cow Rosie (1997):** Produced human protein-enriched milk (2.4 g/litre) containing **human alpha-lactalbumin**, nutritionally more balanced for human babies than natural cow milk.
   - **Biopiracy Case:** In 1997, US company RiceTec was granted patent on Basmati rice cross-bred with semi-dwarf varieties, misappropriating traditional Indian farmers' strains.